RAMCO INSTITUTE OF TECHNOLOGY

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Department of Information Technology

Academic Year: 2024 - 2025(Even Semester)

Degree, Semester & Branch: B. Tech, IV & IT

Course Code & Title: CS3492 & Database Management Systems

Name of the Faculty member: A. Alagulakshmi, AP/IT

Innovative Practice Description

- **Unit / Topic: Unit III/ Deadlock Handling**
- **Course Outcome:** CO3
- **Activity Chosen:** Flipped class Room
- **Justification:**

  In a flipped classroom, students work together to review course materials, including books and videos, at home while taking notes according to their own comprehension. They then put this information to use in the classroom through assignments, debates, and problem-solving exercises. This approach promotes active engagement and self-directed learning.
- **Time Allotted for the Activity: 15 Minutes**
- **Details of the Implementation:**

  One teaching method that flips the conventional learning environment is called "flipped classroom." This method lets students learn at their own speed by delivering direct education outside of the classroom, usually through texts, recorded video lectures, or other materials. This approach keeps students interested and improves their comprehension by giving them more control over their education while allowing teachers to use class time for debate, problem-solving, and active learning. I created a series of questions pertaining to the discussed material and used them in the following class period to gauge their comprehension. Pupils were instructed to respond in class either in writing or by presentation. Student Saravana Kumar presents his self-examined comprehension of deadlock handling in Figure 1. Students are encouraged to actively connect with the subject and solve problems on their own with this technique.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO10	PO11	PSO1
CO3	3	2	3	2	1	2	1	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovate practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO10**	**PO11**	**PSO1**
	3	2	3	2	1	2	1	2
Justification for correlation	Utilize your foundational understanding of concurrency control. Using recuperation techniques to address real-world issues	Determine whether recovery and concurrency control methods are necessary for creating apps.	Capable of creating solutions for any application that uses concurrency control methods and locking protocols	Capable of analyzing stalemate occurrences and recovery mechanisms to draw reliable findings	Utilize research-based expertise, such as recovery mechanisms and concurrency management, to produce reliable results.	Compile the functional and non-functional needs for recovery and concurrency control systems into a report, graphic, or presentation.	As a team member and leader, exhibit your expertise and comprehension of concurrency control and recovery procedures to manage projects.	Exhibit knowledge in concurrency control and recovery mechanisms to provide solutions for complex computational problems.

- **Images / Screenshot of the practice:**

Fig 1: Saravana Kumar S presentation about Deadlock Handling

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***

  - Student felt good, since they can study at their own pace/time.
  - They felt that through such learning, they can explore more.

- ***Benefit of the practice:***
  - More one-to-one time between teacher and student.
  - More collaboration time for students.
  - Students learn at their own pace.
  - Practical things – like missing class due to illness – become less problematic.
  - It encourages students to come to class prepared.

- ***Challenges faced in implementation:***

  - The depth of the subject can be dictated by the student themselves or the group the student isworking with.
  - The time and effort required from a teacher's perspective initially when creating the flipped class material is higher than for a traditional class.

References:

- https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-what-why-and-how- to-implement-a-flipped-classroom-model
- https://en.wikipedia.org/wiki/Flipped_classroom
- https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162- what-why-and-how-to-implement-a-flipped-classroom-model

Signature of Faculty Member **HOD**